Ch.5 Starting the Process

[Step 1: Define mission and objectives]

- Start by defining the result or final output
- Interview – very important because it supports communication

Good interview practices for participants

1. Explain purpose➔ people will be more engaged
2. Thank their contribution, “your feedback is valuable in designing the database”
3. You are the official arbitrator – resolves conflict

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

Good interview practices for interviewer

1. Choose a small number of participants
2. Interview users and managers separately (eliminates conflict)
3. Prepare the questions ahead of time
4. Open-ended questions are recommended and give better results
5. Take notes
6. Give attention to everyone, all must have a turn to speak
7. Keep moving; “we table Q3 and move to Q4”
8. Maintain control
   - re-direct discussion of participants going off-topic
   - explain to dominant voices that your duty is to get input from all

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

A) Define the Mission Statement

- Succinct, no specifics
- Is there a candle at the end of the tunnel
- Good ex. [why?]: *The purpose of the New Starz Talent Agency database is to **maintain** the data we generate, and to **supply information** that supports the **engagement services** we provide to our **clients** and the **management services** we provide to our **entertainers**.*
- Bad ex. [why?]: *The purpose of the Whatcom County Hearing Examiner's database is to keep track of applications for land use, maintain data on applicants, keep a record of all hearings, keep a record of all decisions, keep a record of all appeals, maintain data on department employees, and maintain data for general office use.*

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

Composing a Mission Statement

- Must interview the owner
- Must understand the organization as well as you can

- Examples of questions – some are repeated in a reworded way, why? [Pair the similar questions]
  1. How would you describe the purpose of your organization to a new client?
  2. What would you say is the purpose of your organization?
  3. What is the major function of your organization?
  4. How would you describe what your organization does?
  5. How would you define the single most important reason for the existence of your organization?
  6. What is the main focus of your organization?

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

Case study: Mike's Bikes

- Small suburban bike shop

1. How would you describe the purpose of your organization to a new client?
2. What would you say is the purpose of your organization?
3. What is the major function of your organization?
4. How would you describe what your organization does?
5. How would you define the single most important reason for the existence of your organization?
6. What is the main focus of your organization?

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

Zach interviews Mike

- ZACHARY: "Can you tell me why you believe you need a database?"
- MIKE: "I think we need a database just to keep track of all our inventory. I'd also like to keep track of all our sales as well."
- ZACHARY: "I'm sure the database will address those issues. Now, what would you say is the single most important function of your business?"
- MIKE: "To provide a wide array of bicycle products and bicycle-related services to our customers. We have a lot of great customers. And regular ones too! They're our biggest asset."

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

Resulted mission statement

ZACHARY: "Can you tell me why you believe you need a database?"

MIKE: "I think we need a database just to keep track of all our **inventory**. I'd also like to keep track of all our **sales** as well."

ZACHARY: "I'm sure the database will address those issues. Now, what would you say is the single most important function of your business?"

MIKE: "To provide a wide array of bicycle **products** and bicycle-related **services** to our customers. We have a lot of great customers. And regular ones too! They're our biggest asset."

- Mission statement: The purpose of the Mike's Bikes database is to **maintain the data** we need to support our **retail sales** business and our **customer service** operations.

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

B) Define the Mission Objectives (=tasks)

- Succinct and unambiguous
- Very important, help create tables, fields, views, data integrity, business rules
- Must interview users and managers
- Pay attention to answers!!!
  - Explicit information
  - Implicit information – “read between the lines”

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

Questions for finding objectives

1. What kind of work do you perform on a daily basis?
2. How would you define your job description?
3. What kind of data do you work with?
4. What types of reports do you generate?
5. What types of things do you keep track of?
6. What types of services does your organization provide?
7. How would you describe the type of work you do?
   - *Possible answer to this question: "First, I try to determine the general problem with the **vehicle**. Then I fill out a **work order** and note my assessment of the problem. Finally, I send the vehicle to the next available **service team**." ➔ see objective on next slides*

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

Mike's Bikes Objectives

1. Maintain information on customer **vehicles**.
2. Keep track of **work orders**.
3. Maintain information on our **service** teams.
4. Maintain information on our mechanics.
5. Maintain information on our customers.

[Last two are derived from hidden info.]

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.

A few additional resulted objectives

ZACHARY: "Can you give me an idea of the things you'd like to track in the database?"

MIKE: "Oh sure, that's pretty easy. I want to keep track of our inventory, our customers, and our sales."

ZACHARY: "Is there anything else that you can think of that is related to these subjects?"

MIKE "Well, I guess if we're going to keep track of our inventory, we should know who our suppliers are."

ZACHARY: "What about the sales reps involved in each sale?"

MIKE: "Oh yeah, we should definitely keep information about our employees. If nothing else, it's a good idea to do this from a human resources point of view. At least, that's what my wife tells me!"

Etc. ….

1. Maintain complete **inventory** information.
2. Maintain complete **customer** information.
3. Track all customer **sales**.
4. Maintain complete **supplier** information.
5. Maintain complete **employee** information.

Finally, review objectives with Mike and staff, then save them for next step!

From *Database Design for Mere Mortals®, Third Edition*, by Michael J. Hernandez (ISBN: 9780321884497). Copyright 2013 by Michael J. Hernandez. All rights reserved.